

An Internship Report
Of
Cybersecurity Virtual Internship

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By

Roll Number of Student: 2410031294

Name of Student: DIPANSHU
KUMAR

Batch: 2024-2028

CANDIDATE'S DECLARATION

I, DIPANSHU KUMAR, do hereby declare that the internship report titled “Cybersecurity Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references and sources used in this report have been appropriately acknowledged and cited. I affirm that this report is my original work and has not been submitted to any other institute or university for the fulfilment of any degree or diploma requirement. I further declare that I shall be solely responsible for any kind of copyright violation or plagiarism arising in connection with this report.

Signature:-

Student Name: DIPANSHU KUMAR

Roll No.: 2410031294

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their inspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the **AICTE–EduSkills Virtual Internship Program** team and **EduSkills Academy**, with the support of **Palo Alto Networks**, for providing me with the opportunity and structured platform to learn and gain practical exposure in the domain of **Cybersecurity**. I am also thankful to my institute, **IILM University, Greater Noida**, and the faculty of the **School of Computer Science and Engineering** for their continuous support and encouragement.

I express my gratitude to my parents for their blessings, constant support, and encouragement throughout the Internship Program.

I would also like to express my sincere appreciation to everyone who directly or indirectly contributed to the successful completion of this internship and report.

Signature:-

Student Name: DIPANSHU KUMAR

Roll No.: 2410031294

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3. INTERNSHIP COMPLETION CERTIFICATE

**MINISTRY OF
EDUCATION**

**AICTE**
All India Council of Technical Education

**NATIONAL
INTERNSHIP
PORTAL**

**EduSkills®**
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

Dipanshu Kumar

IILM University, Greater Noida

has successfully completed the 10-week
Cybersecurity Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

Supported By  **paloalto®**
NETWORKS
Saravanan Rajagopal
Training Partner Manager, APAC
Palo Alto Networks
Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education
Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkillsCertificate ID : 9b5b994248741a89905398d80b99e891
Student ID : STU69025ec384e4d1761763011

GRADE: O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | F (Pass): 30-39 | F (Fail): Below 30

INTERNSHIP OFFER LETTER

Reference No.: [YGSDYG5EUG5](#)

Student Name	Dipanshu Kumar
Roll No.	2410031294
Stream	B.Tech
Branch	B.Tech (Computer Science and Engineering – Cyber Security)
Institute Name	IILM University, Greater Noida
Internship Domain	Google AI-ML
Internship Duration	June 2026 – August 2026
AICTE Student ID	2410031294

We are pleased to present this SAMPLE offer letter for Dipanshu Kumar for an 8-Week AICTE–EduSkills Virtual Internship Program in the Google AI-ML domain.

This document is a sample/mock-up prepared for demonstration and academic formatting purposes. It does not represent an actual offer, selection, or official credential.

The sample internship is designed to provide structured learning combined with practical exposure in Machine Learning and Deep Learning, with emphasis on TensorFlow, computer vision, image classification, object detection, and mobile application integration.

Internship Structure & Evaluation:

- The internship follows a week-wise structured learning plan.
- Weekly assessments are used to track progress and understanding.
- Selected modules may include project documentation and assignments.
- A final assessment may be conducted during the final week.
- The overall evaluation may consider assessments, project submissions, and the final test.

Sample Completion Statement: Upon successful completion of the required learning activities and assessments, a virtual internship certificate may be awarded by the relevant program authority.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the **10-week Cybersecurity Virtual Internship** undertaken under the **AICTE–EduSkills Virtual Internship Program**, in the domain of **Cybersecurity**, supported by **Palo Alto Networks**. The internship was carried out from **June 2026 to August 2026** as part of the requirement of the **Bachelor of Technology (Computer Science and Engineering)** programme at **IILM University, Greater Noida**. The primary aim of this internship was to develop a strong foundational understanding of cybersecurity concepts and gain practical exposure to various aspects of cybersecurity, including security threats, vulnerabilities, network security, information security, and modern security practices. The internship followed a structured learning approach combining conceptual understanding with practical learning, assessments, and internship-related activities. The program provided an opportunity to enhance technical knowledge, develop problem-solving skills, and understand the practical importance of cybersecurity in protecting digital systems and information. On successful completion of the internship requirements, a **Certificate of Virtual Internship** was awarded to me by **EduSkills**, with the internship supported by **Palo Alto Networks** and associated with **AICTE and the Ministry of Education**.

Organization Profile

EduSkills Academy is an ISO 9001:2015 and ISO/IEC 20000-1:2018 certified organization working under the theme “Nation Building Through Skills”. EduSkills partners with the All India Council for Technical Education (AICTE) to run the AICTE – EduSkills Virtual Internship Program under the patronage of the AICTE National Internship Portal. For the Google AI-ML track, EduSkills collaborates with Google for Developers to design an industry-aligned curriculum, delivered through the Google Developer Program’s learning pathways, quizzes, and skill badges, that helps engineering students build practical machine-learning and computer-vision capability.

4.2 Problem Statement

Although a large number of engineering students learn the theoretical foundations of cybersecurity during their academic coursework, very few get structured, hands-on exposure to understanding and applying cybersecurity concepts in practical scenarios. Concepts such as network security, cyber threats, vulnerabilities, information security, and

security practices are often understood primarily at a theoretical level, with limited opportunities to gain practical exposure to real-world cybersecurity challenges. The problem addressed by this internship was to develop a practical understanding of cybersecurity fundamentals and apply the knowledge gained through structured learning, assessments, and internship-related activities, while gaining awareness of modern cybersecurity threats, security mechanisms, and approaches used to protect digital systems and information. Project Objectives

4.3 Scope of the Project

The scope of this internship encompasses a structured and practical introduction to the field of **Cybersecurity**, beginning with the understanding of fundamental cybersecurity concepts and extending to the identification of common cyber threats, vulnerabilities, security risks, and methods used to protect digital systems and information. The internship provided exposure to important areas of cybersecurity, including **network security, information security, threat awareness, vulnerability management, access control, security monitoring, and basic approaches to preventing and responding to cyber incidents**. It also focused on developing an understanding of how security principles can be applied to modern computing environments and how organizations can reduce the risks associated with cyberattacks and unauthorized access.

The internship further covered the importance of maintaining the **confidentiality, integrity, and availability (CIA)** of information and understanding the role of security mechanisms in protecting networks, applications, devices, and digital resources. Through structured learning activities, practical exercises, assessments, and internship-related tasks, the program helped bridge the gap between theoretical cybersecurity concepts studied during academic coursework and their practical relevance in real-world situations. The scope also included developing analytical and problem-solving skills required to recognize potential security issues, understand their impact, and consider appropriate security measures for mitigating associated risks.

4.4 Technologies and Tools Used

- **Cybersecurity Fundamentals:** Cybersecurity concepts, information security, security principles, and risk awareness
- **Network Security:** Network protection concepts, network security mechanisms, and secure communication
- **Threat & Vulnerability Management:** Cyber threats, vulnerabilities, attack awareness, risk identification, and basic mitigation strategies
- **Security Operations:** Security monitoring, threat detection, incident awareness, and basic incident-response concepts
- **Access & Information Security:** Authentication, authorization, access control, data protection, and information security practices
- **Security Platforms & Learning:** Palo Alto Networks cybersecurity learning resources, practical learning activities, assessments, and internship-related exercises

3.1 • System Architecture

The general architecture followed during the internship for the Google AI-ML learning track can be summarized in the following layers:

- **Model Layer:** A convolutional neural network built and trained using TensorFlow forms the core intelligence layer for image recognition and classification.
- **Conversion Layer:** Trained models are converted into a lightweight, on-device format using TensorFlow Lite and the TFLite Model Maker.
- **Detection Layer:** Pretrained and custom-trained object detectors identify and localize objects within static images and live camera feeds.
- **Search Layer:** A product image search feature combines on-device object detection with backend integration to enable visual product search.
- **Classification Layer:** A custom image classifier further categorizes detected objects/images with improved accuracy.
- **Application Layer:** The trained and converted models are integrated into a mobile application, consuming the detection, search, and classification capabilities.

- Assessment Layer: Weekly quizzes, learning-pathway badges, and a Final Credential Validation / Internship Grade Point Assessment track progress and final grading.

3.2 Methodology

The internship followed a structured, week-wise methodology combining conceptual learning, hands-on labs, and quiz-based assessments, culminating in a Final Credential Validation, as summarized in the table below.

Week	Module	Description
1	Program Neural Networks with TensorFlow	Foundational concepts of machine learning, from first principles through to building convolutional neural networks (CNNs) for advanced image recognition and classification.
2	Get Started with Object Detection	Basics of object detection, and integrating pretrained object detectors into mobile applications.
3	Go Further with Object Detection	Training custom object-detection models using TensorFlow Lite and the TFLite Model Maker.
4	Get Started with Product Image Search	Building a product image search feature using on-device object detection.
5	Get Started with Product Image Search	Detecting objects in static images and live camera feeds for visual product search.
6	Go Further with Product Image Search	Building backend integration for a product image search mobile application.
7	Go Further with Image Classification	Building custom image-classification models and improving classification skill.
8	Go Further with Image Classification	Creating and integrating a custom image-classifier model into an application.
Final	Final Credential Validation	Final Credential Validation / Internship Grade Point Assessment.

Note: There was no stipend associated with this internship.

Each week concluded with quiz-based learning-pathway assessments and skill badges to evaluate conceptual understanding. In the final phase, a Final Credential Validation / Internship Grade Point Assessment was conducted, and the overall grade was based on performance across the weekly modules and this final assessment.

3.3 Expected Outcomes

- Ability to explain and apply the first principles of machine learning to build convolutional neural networks using TensorFlow.

- Practical understanding of object detection, including the use of pretrained detectors and custom-trained TensorFlow Lite models.
- Hands-on exposure to building a product image search feature combining on-device detection with backend integration.
- Capability to build, evaluate, and improve custom image-classification models.
- Skill to integrate trained and converted TensorFlow Lite models into a mobile application.
- A completed set of weekly learning-pathway modules and skill badges, along with successful clearance of the Final Credential Validation / Internship Grade Point Assessment, resulting in an “Outstanding” grade.

3.4 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate issued by EduSkills, confirming the successful completion of the 10-week Cybersecurity Virtual Internship under the AICTE–EduSkills Virtual Internship Program, are attached in Chapter 3 of this report. The internship was supported by Palo Alto Networks and conducted during June–August 2026. The certificate issued upon successful completion serves as official documentation of the internship and is included as supporting evidence in this report.

4. BIBLIOGRAPHY/REFERENCES

1. □ **EduSkills Academy**, “AICTE–EduSkills Virtual Internship Program – Cybersecurity,” Internship Program, 2026.
2. □ **EduSkills Academy**, “Certificate of Virtual Internship – Cybersecurity,” Certificate ID: 9b5b994248741a69905398d60b99e891, Student ID: STU69025ec364e4d1761763011, 2026.
3. □ **AICTE National Internship Portal**, “AICTE–EduSkills Virtual Internship Program,” All India Council for Technical Education, Ministry of Education, Government of India.
4. □ **Palo Alto Networks**, “Cybersecurity Academy,” Palo Alto Networks.
5. □ **Palo Alto Networks**, “Cybersecurity Learning and Training Resources,” Palo Alto Networks.
6. □ **EduSkills Academy**, “AICTE–EduSkills Virtual Internship Program,” EduSkills Foundation.
7. □ **All India Council for Technical Education (AICTE)**, “National Internship Portal,” Ministry of Education, Government of India.
8. □ **IILM University**, “School of Computer Science and Engineering,” IILM University, Greater Noida.